

- RAG Learning

2.6 Sequences

AQA GCSE Mathematics 8300 — Algebra — spec refs A23–A25 — Papers 1 (non-calculator), 2 and 3 (calculator)

2.6.1 Generating Sequences (A23)

A **sequence** is an ordered list of numbers called **terms**. The **position** of a term is usually called n : the 1st term has $n = 1$, the 2nd term $n = 2$, and so on.

Term-to-term rules

A **term-to-term rule** tells you how to get from one term to the next. You need a starting term as well as the rule.

Worked Example: The first term of a sequence is 6. The rule is "multiply by 3 then subtract 4". Write down the next three terms.

$$6 \times 3 - 4 = 14, \quad 14 \times 3 - 4 = 38, \quad 38 \times 3 - 4 = 110$$

Sequence: 6, **14, 38, 110**

Term-to-term rules can be written with **suffix notation**: u_n is the n th term and u_{n+1} is the term after it. So "multiply by 2 then subtract 3, starting at 5" is written $u_{n+1} = 2u_n - 3$, $u_1 = 5$, giving 5, 7, 11, 19, ... Any unusual recursive rule like this will always be defined in the question.

Position-to-term rules (the n th term)

A **position-to-term rule** gives any term directly from its position. Substitute $n = 1, 2, 3, \dots$

Worked Example: Write down the first three terms and the 20th term of the sequence with n th term $5n - 2$.

$$n = 1: 5 - 2 = 3; \quad n = 2: 10 - 2 = 8; \quad n = 3: 15 - 2 = 13$$

$$n = 20: 5 \times 20 - 2 = \mathbf{98}$$

Worked Example: Write down the first three terms of the sequence with n th term $n^2 + 4$.

$$1 + 4 = 5, \quad 4 + 4 = 8, \quad 9 + 4 = 13 \rightarrow \mathbf{5, 8, 13}$$

Sequences from patterns and diagrams



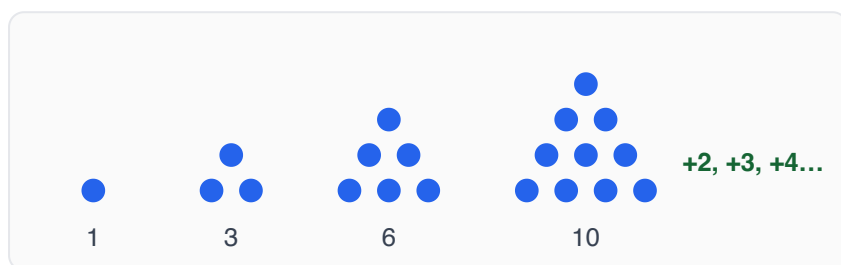
Each new square needs 3 more sticks, so the number of sticks goes 4, 7, 10, ... with n th term $3n + 1$. Pattern 50 needs $3 \times 50 + 1 = 151$ sticks.

2.6.2 Special Sequences (A24)

Sequence	First terms	n th term
Square numbers	1, 4, 9, 16, 25, 36, ...	n^2
Cube numbers	1, 8, 27, 64, 125, ...	n^3
Triangular numbers	1, 3, 6, 10, 15, 21, ...	$\frac{n(n+1)}{2}$
Even numbers	2, 4, 6, 8, ...	$2n$

Odd numbers	1, 3, 5, 7, ...	$2n - 1$
Powers of 2	2, 4, 8, 16, ...	2^n

Triangular numbers



Each triangle adds a new bottom row one dot longer than the last, so the differences are 2, 3, 4, 5, ... The 12th triangular number is $\frac{12 \times 13}{2} = 78$.

Arithmetic progressions

An **arithmetic** (or **linear**) sequence increases or decreases by the same amount each time. This amount is the **common difference** d . Example: 20, 17, 14, 11, ... has $d = -3$.

Fibonacci-type sequences

In a **Fibonacci-type** sequence, each term is the **sum of the two previous terms**. The Fibonacci sequence itself is 1, 1, 2, 3, 5, 8, 13, 21, ... but any two starting numbers can be used.

If the first two terms are a and b , the sequence is $a, b, a + b, a + 2b, 2a + 3b, 3a + 5b, \dots$. This lets you form equations.

Worked Example: A Fibonacci-type sequence has 3rd term 11 and 5th term 29. Find the first two terms.

3rd term: $a + b = 11$ 5th term: $2a + 3b = 29$

Double the first: $2a + 2b = 22$. Subtract: $b = 7$, so $a = 4$.

Sequence: **4, 7**, 11, 18, 29 ✓

Geometric progressions

In a **geometric** sequence each term is **multiplied** by the same number, the **common ratio** r . Find r by dividing a term by the one before it. At Foundation r is a positive rational number, e.g. 2, 3, $\frac{1}{2}$, 1.5.

Geometric sequence with first term a and common ratio r :

$$a, ar, ar^2, ar^3, \dots \quad n\text{th term} = ar^{n-1}$$

Worked Example: A geometric sequence has first term 5 and a positive common ratio. The 3rd term is 45. Find the 6th term.

$$5r^2 = 45 \Rightarrow r^2 = 9 \Rightarrow r = 3$$

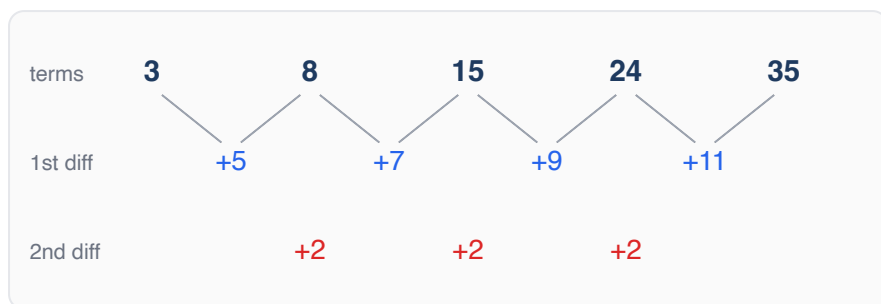
Terms: 5, 15, 45, 135, 405, **1215**

Worked Example: A geometric sequence begins 64, 16, ... Find the next two terms.

$$r = 16 \div 64 = \frac{1}{4}, \text{ so the next terms are } 16 \times \frac{1}{4} = \mathbf{4} \text{ and } \mathbf{1}.$$

Quadratic sequences

A **quadratic** sequence has first differences that change, but **second differences** that are constant. Its n th term contains n^2 as the highest power.



The second difference is constant (2), so 3, 8, 15, 24, ... is quadratic. Continuing the pattern of first differences (+11, +13) gives the next terms 35 and 48.

Exam Tip: To decide what type a sequence is: constant first difference $\Rightarrow$ arithmetic; constant second difference $\Rightarrow$ quadratic; constant ratio $\Rightarrow$ geometric; each term = sum of previous two $\Rightarrow$ Fibonacci-type.

Other sequences Higher

At Higher tier, geometric sequences can have a **surd** common ratio or a **negative** ratio, and you may meet other sequences defined in the question.

Worked Example: A geometric sequence begins 3, $3\sqrt{2}$, 6, ... Find the 9th term.

$$r = \frac{3\sqrt{2}}{3} = \sqrt{2}$$

$$9\text{th term} = 3 \times (\sqrt{2})^8 = 3 \times 2^4 = 3 \times 16 = 48$$

Worked Example: A geometric sequence has first term 2 and second term $2\sqrt{3}$. Which is the first term greater than 500?

$$r = \sqrt{3}, \text{ and every two steps multiply by } (\sqrt{3})^2 = 3: \text{ terms } 2, 2\sqrt{3}, 6, 6\sqrt{3}, 18, \dots$$

$$11\text{th term} = 2 \times 3^5 = 486; 12\text{th term} = 486\sqrt{3} \approx 841.8. \text{ So the } \mathbf{12\text{th}} \text{ term.}$$

A negative ratio makes the signs alternate: 5, -10 , 20, -40 , 80, ... has $r = -2$.

2.6.3 The n th Term of a Linear Sequence (A25)

For a linear sequence with common difference d :

$$n\text{th term} = dn + (\text{first term} - d)$$

The value " $\text{first term} - d$ " is the **zeroth term** — the term before the first.

Worked Example: Find the n th term of 7, 11, 15, 19, ...

Common difference 4, so start with $4n$: 4, 8, 12, 16, ...

Each term is 3 more than $4n$, so the n th term is **$4n + 3$** .

Worked Example: Find the n th term of 40, 37, 34, 31, ... and the 25th term.

$d = -3$; zeroth term $= 40 + 3 = 43$. n th term $=$ **$43 - 3n$**

25th term $= 43 - 75 =$ **-32**

Is a number in the sequence?

Set the n th term equal to the number and solve. It is a term only if n is a **positive whole number**.

Worked Example: Is 150 a term of $7n + 3$? $7n + 3 = 150 \Rightarrow 7n = 147 \Rightarrow n = 21$.

Yes, it is the 21st term.

Is 98 a term of $4n + 4$? $4n = 94 \Rightarrow n = 23.5$. Not a whole number, so **no**.

Problems with linear sequences

Worked Example: Sequence A has n th term $5n - 4$. Sequence B begins 100, 96, 92, ... At which position are the terms equal?

B: $d = -4$, zeroth term 104, so n th term $= 104 - 4n$.

$5n - 4 = 104 - 4n \Rightarrow 9n = 108 \Rightarrow n = 12$. Both 12th terms equal **56**.

Worked Example: 1 table seats 6, 2 tables seat 10, 3 tables seat 14. What is the least number of tables for 75 people?

n th term $4n + 2$. Need $4n + 2 \geq 75 \Rightarrow n \geq 18.25$, so **19** tables (18 tables seat only 74).

Exam Tip: The coefficient of n is the common difference, *not* the first term. A common error is writing 3, 7, 11, 15 as " $n + 4$ ". Always check your formula by substituting $n = 1$ and $n = 2$.

2.6.4 The n th Term of a Quadratic Sequence Higher

For a quadratic sequence with n th term $an^2 + bn + c$:

$$a = \frac{\text{second difference}}{2}$$

1. Find the second difference and halve it to get a .
2. Subtract an^2 from each term.
3. Find the n th term of the remaining linear sequence ($bn + c$).
4. Combine: $an^2 + bn + c$, and check with $n = 1$.

Worked Example: Find the n th term of 3, 10, 21, 36, 55, ...

First differences: 7, 11, 15, 19 Second difference: 4, so $a = 2$.

n	1	2	3	4	5
Term	3	10	21	36	55

$2n^2$	2	8	18	32	50
Term $-2n^2$	1	2	3	4	5

The remainder is n , so the n th term is $2n^2 + n$. Check: $n = 1: 2 + 1 = 3 \checkmark$

Worked Example: Find the n th term of 12, 17, 20, 21, 20, ...

First differences: 5, 3, 1, -1 Second difference: -2 , so $a = -1$.

Term $-(-n^2)$, i.e. term $+n^2$: 13, 21, 29, 37, 45 $\Rightarrow 8n + 5$

n th term $= -n^2 + 8n + 5$. Check: $n = 2: -4 + 16 + 5 = 17 \checkmark$

Worked Example: The n th term of a sequence is $n^2 + 2n$. Show that 120 is a term.

$$n^2 + 2n = 120 \Rightarrow n^2 + 2n - 120 = 0 \Rightarrow (n + 12)(n - 10) = 0$$

$n = 10$ (reject $n = -12$ as positions are positive). 120 is the **10th** term.

Exam Tip: Halve the second difference — using the second difference itself as the coefficient of n^2 is the most common mistake. If the second difference is negative, the n^2 term is negative too.

Quick Reference Summary

Sequence type	How to spot it	n th term	Tier
Arithmetic (linear)	Constant first difference d	$dn + (\text{first term} - d)$	Foundation
Geometric	Constant ratio r (term $\div$ previous)	ar^{n-1}	Foundation (r positive rational); Higher (r surd or

	term)		negative)
Quadratic	Constant second difference	$an^2 + bn + c$, a = half second difference	Recognise: Foundation; n th term: Higher
Fibonacci-type	Each term = sum of previous two	Use $a, b, a + b, a + 2b, \dots$	Foundation
Square / cube / triangular	1, 4, 9... / 1, 8, 27... / 1, 3, 6...	$n^2 / n^3 / \frac{n(n+1)}{2}$	Foundation